Intersection of Race and Socioeconomic Status on Concussion Recovery among NCAA Student-Athletes: A CARE Consortium Study

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ABSTRACT

BOLTZ, A. J., A. K. MEMMINI, B. L. BRETT, T. R. SNEDDEN, A. M. YENGO-KAHN, A. CHANDRAN, D. P. CONWAY, R. SHAH, P. F. PASQUINA, T. W. MCALLISTER, M. A. MCCREA, C. L. MASTER, and S. P. BROGLIO, CARE CONSORTIUM INVESTIGATORS. Intersection of Race and Socioeconomic Status on Concussion Recovery among NCAA Student-Athletes: A CARE Consortium Study. *Med. Sci. Sports Exerc.*, Vol. 55, No. 12, pp. 2180–2193, 2023. **Purpose:** The objectives of this study are to 1) describe collegiate student-athlete (SA) race and household income and 2) evaluate time to normal academic performance (i.e., return to learn (RTL)), initiation of the return to play (iRTP) protocol, RTP protocol duration, and time to unrestricted RTP (URTP) after sustaining sport-related concussion (SRC). **Methods:** Data were collected between 2014 and 2020 by the Concussion Assessment, Research, and Education Consortium. Baseline data were used to characterize participant demographics (N = 22,819) and post-SRC outcomes (n = 5485 SRC) in time to RTL (n = 1724) and RTP outcomes (n = 2646) by race. Descriptive statistics and nonparametric tests examined differences across race by demographic and injury characteristics. Kaplan–Meier curves estimated median days to RTL, iRTP protocol, RTP protocol completion, and URTP by race and covariate measures. Multivariable Cox proportional hazards regression assessed the effect of race on risk of RTL and RTP recovery time points. **Results:** SA largely identified as White (75%) followed by Black (14%), multiracial (7%), and Asian (3%). More than half (53%) of all SA reported a household income of >$120,000, whereas 41% of Black SA reported a household income <$60,000. Race was not associated with relative risk of RTL or iRTP but was associated with RTP protocol completion and URTP. Non-Black/non-White SA were 17% less likely (adjusted hazard ratio = 0.83; 95% confidence interval = 0.71, 0.97) to complete the RTP protocol, and Black SA were 17% more likely (adjusted hazard ratio = 1.17; 95% confidence interval = 1.05, 1.31) to reach the URTP time point compared with White SA. **Conclusions:** The present findings suggest collegiate SA enrolled in the Concussion Assessment, Research, and Education Consortium are primarily White and come from household incomes well above the US median. Race was not associated with RTL or iRTP but was associated with RTP protocol duration and total time to URTP. Clinicians should be conscientious of how their implicit or preconceived biases may influence SRC management among National Collegiate Athletic Association SA. **Key Words:** MILD TRAUMATIC BRAIN INJURY, CLINICAL MANAGEMENT, SOCIAL CONSTRUCT, BIOPSYCHOSOCIAL, RETURN TO LEARN, RETURN TO PLAY

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Sport-related concussion (SRC) has been characterized as a biopsychosocial injury (1). Understanding each pillar of the biopsychosocial SRC complex in relation to the individual has been a targeted focus in recent position and consensus documents (2,3). SRC biology is characterized as an acute disruption of ionic and metabolic processes of the cerebral tissue following a direct or indirect force to the head, neck, or body (4). These neuronal disturbances commonly result in well-documented signs and symptoms, along with motor and cognitive impairments (4) that resolve spontaneously within 4 wk for most collegiate student-athletes (SA) (5). Noninjury factors, such as pre- and postinjury personality and

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psychological traits, are known to influence SRC risk (6) and recovery (7). Furthermore, social factors, such as personal identity(ies) and team culture, have been associated with injury reporting behaviors (8,9).

Within the biopsychosocial framework, race is defined as a social construct (10). In 2020, the US Census reported the three most prevalent races as White (61.6%), Black (12.4%), and Asian (6%) (11), which are tightly linked to socioeconomic status (SES). The American Psychological Association defines SES as a social class status measured by household income, education, and occupation status (12). Within the United States, those who identify as Asian are among the most likely to obtain a bachelor's degree or higher (55.6%), followed by White (36.9%) and Black (22.5%) (11). Median household incomes follow a similar trend, with the highest median incomes reported by Asian households, followed by White and Black (11). Beyond educational attainment and financial security, race and SES often carry unique perspectives and life experiences, including racial discrimination, marginalization, and microaggressions, which may influence SRC knowledge (13), reporting (13), access to healthcare (14), and injury outcomes (15). Indeed, contemporary literature appraising the relationship between race and SRC recovery has resulted in heterogeneous findings across the intersections of age, sex, and sport contact level(s) (15–20).

Furthermore, health outcomes are influenced by a complex interplay among social constructs and their determinants, such as SRC knowledge (13,21), injury disclosure behaviors (22), and recovery trajectories (15). Indeed, premorbid and concomitant psychological factors are known to influence recovery trajectories (16,23), and implicit biases from healthcare providers on the basis of patients' race may influence postinjury management (24), including return to learn (RTL) and return to play (RTP). Despite advancements in SRC evaluation and management over the past two decades (25), a holistic appraisal of biopsychosocial factors and their relationship with SRC recovery is lacking (2).

Therefore, the objectives of this investigation were to 1) comprehensively characterize the diversity of household incomes and racial characteristics of collegiate SA enrolled in the Concussion Assessment, Research, and Education (CARE) Consortium across sports; 2) assess the relationship of race and time to RTL; and 3) determine if there is a relationship between race and time to initiation of the RTP (iRTP) protocol, RTP protocol duration, and time to unrestricted RTP (URTP) following SRC.

METHODS

The National Collegiate Athletic Association (NCAA) and Department of Defense (DoD) sponsored the CARE Consortium, which is a 30-site investigation (26 civilian institutions and four military service academies) designed to assess the natural history of SRC across collegiate varsity athletes and military cadets/midshipmen in the United States. A full description of the methods has been previously described in detail elsewhere (26).

Briefly, each consenting participant completed a baseline evaluation at the time of enrollment, including sex, primary sport, race (US Census categories: White, Black, and non-Black/non-White (Hawaiian Islander, multiple races, Asian, Alaskan Indian)), total parent(s)/guardian(s) income extracted from the *Hollingshead* Four-Factor Index of SES (27), as well as symptoms using the Sport Concussion Assessment Tool-3 (SCAT3) (28,29). In order to preserve statistical power, ethnicity was not considered because of a low sample size of Hispanic and/or Latinx SA within the dataset. Post-SRC assessments captured injury and recovery characteristics (e.g., altered mental status, initial symptom burden via SCAT3, categories of academic supports provided, etc.). Assessment measures were repeated immediately following an SRC and at multiple time points throughout the recovery process. Institutional review board approval was obtained from each participating site, and the US Military Human Research Protection Office and each participant provided written informed consent. This study was conducted in accordance with the Declaration of Helsinki.

Participant Sample

Between 2014 and 2020, a total of 47,860 participants completed at least one baseline evaluation. As this investigation was intended to explore the relationship between race and SRC metrics, military service academy (MSA) members not participating in a varsity level sport (n = 13,695) and those with incomplete demographics information (e.g., missing race and/or household income, n = 11,346) were removed from the dataset. In instances where an SA had baseline evaluations in subsequent years, only the initial baseline information was used. A total of 22,819 NCAA SA were included in the dataset (Fig. 1). Among the NCAA SA, 5485 SRC were documented by the local medical teams using a common SRC definition (30) and a standardized assessment battery (26) to inform the diagnosis and postinjury management. SRC were excluded if they were missing both postinjury and 24- to 48-h postinjury data (n = 190) or if they were not the first SRC during the study period (n = 681), resulting in 4614 SRC. Please see Figure 1 for details regarding the inclusion and exclusion criteria for each of the following analyses.

Race and SES among Collegiate SA

The association between race and SES on the 22,819 NCAA SA was examined by self-reported sex, primary sport, race, and median household income using descriptive statistics (counts and frequencies). The distribution (counts (%)) of primary sport by race, primary sport by median household income, median household income and race, and primary sport by race with average median income household was also evaluated.

Time to Return to Normal Academic Performance (i.e., RTL)

To understand the relationship between race and RTL, SRC were omitted from the analyses if they were missing RTL data (n = 833) or if the SA reported 0 d to RTL (n = 2). Additional

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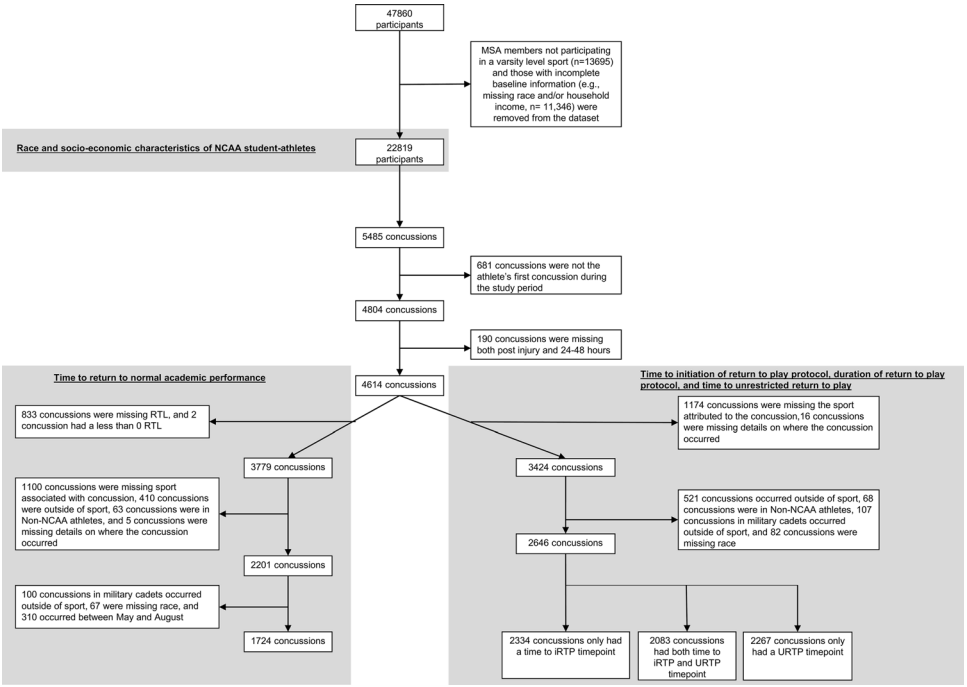

FIGURE 1—CONSORT diagram of participants to evaluate socioeconomic characteristics, RTL, and RTP outcomes.

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exclusion criteria are described in Figure 1. The final dataset included 1724 SRC to assess differences across RTL.

Statistical analysis. To evaluate the time to RTL, demographic and injury data were described using descriptive statistics (counts (frequencies), as well as means (standard deviations [SD]) or medians (interquartile ranges [IQR]), as appropriate) by self-identified race. Kruskal–Wallis tests were used to examine differences across race due to nonparametric distribution among continuous variables. χ^2 tests, or Fisher's exact tests where observations were <5, were calculated to determine differences among categorical variables. The outcome variable of interest was defined as self-reported total time (in days) for the student to return to "normal academic performance" (i.e., RTL) following SRC. Kaplan–Meier estimates (median, 95% confidence intervals (95% CI)) were used to evaluate time to RTL overall, and by sex, NCAA division, contact sport category, and difference between baseline and postinjury SCAT3 symptom severity scores across race. Log-rank tests were used to evaluate differences in RTL across levels of the aforementioned groups.

Lastly, multivariable Cox proportional hazards (adjusted hazard ratios (HR_{Adj}), with 95% CI) models were used to assess the effect of race (Black, non-Black/non-White, White (referent)) (15) on time to RTL adjusting for covariates suggested to influence SRC recovery trajectories including sex (males, females (referent)) (31), SRC history (0 (referent), 1, 2, 3+) (15), sport contact level (contact (referent), limited/noncontact) (32), difference between baseline and postinjury SCAT3 symptom severity score (<20 (referent), ≥20) (16), and median time to asymptomatic (<7 d (referent), ≥7 d). After fitting the full model, we observed the difference in SCAT3 symptom severity scores (baseline subtracted from postinjury symptom severity scores) covariate violated the proportional hazards assumption; therefore, we stratified the sample by this variable and constructed two separate models based on the median difference in symptom severity scores. All other aforementioned covariates and their denoted parameterizations were included in each model. Both models overall, and all covariates did not violate the proportional hazards assumption.

Time to iRTP Protocol, Duration of RTP Protocol, and Time to URTP

Of the 4614 incident SRC, the final dataset included 2464 SRC to assess differences across the RTP time points. Records were excluded from the analyses for specific criteria denoted in Figure 1.

Statistical analysis. Demographic (e.g., sex, NCAA division, NCAA sport category) data collected at baseline and postinjury data were characterized using descriptive statistics (counts (%) or means (SD)/medians (IQR) as appropriate) and examined both within the pooled sample and stratified by race. χ^2 tests, or Fisher's exact tests, were used to examine differential distributions of categorical variables across race. Considering the distributional properties of continuous variables included, Kruskal–Wallis tests were used to examine group differences across race.

We examined three recovery timeframes using univariate and multivariable survival methods. Specifically, univariate Kaplan–Meier estimates (median, 95% CI) were estimated to evaluate the probability of 1) time to iRTP protocol; 2) time to RTP protocol completion; 3) time to URTP by race, and race stratified by biological sex (female, male), contact level of the sport they were participating in at time of SRC (contact, limited contact, noncontact), SRC history (≤2, ≥3), and difference in SCAT3 symptom severity score (<21, ≥21). Log-rank

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tests were used to evaluate differences within each timeframe (iRTP protocol, RTP protocol duration, URTP duration) across aforementioned variables. Time to iRTP protocol data were censored at 30 d (~4% of records were censored), RTP duration data were censored at 20 d (~7% of records were censored), and URTP duration data were censored at 60 d (~2% of records were censored). Censoring in the present study was based on the current understanding of the clinical recovery timeline associated with SRC (5).

Three separate multivariable Cox proportional hazards models were constructed to assess risk of iRTP protocol, RTP protocol completion, and URTP. Covariates were identified *a priori* as potential modifiers of recovery, and in accordance with existing literature (5). All models included SA's race (White (referent), Black, non-Black/non-White) and the following covariates: biological sex (female (referent), male), contact level of the sport they were participating in at time of SRC (contact (referent), limited contact, noncontact) (33), SRC history (≤2 (referent), ≥3), and difference in SCAT3 symptom severity score (<21 (referent), ≥21). Difference in SCAT3 symptom severity score was parameterized based on the median score in the present study.

HR_{Adj} with corresponding 95% CI were estimated. The proportional hazards assumptions for the model overall, and for each covariate, were evaluated to ensure fidelity and stability of the resulting coefficient. The overall proportional hazard assumption for each model was checked by inspecting the log [log(survival)] plots. We used the "zph()" package in R to check for the proportionality assumption to examine the Schoenfeld residuals against the transformed time ($P < 0.05$ indicated time-dependent coefficients).

In the time to iRTP protocol and URTP models, the proportional hazards assumption was violated for the SCAT3 symptom severity and sport contact level variables; therefore, we constructed a time-varying covariate using quartiles of time (in days) to iRTP protocol (Timegroup = 1 (0.1, 3.9); Timegroup = 2 (3.9, 6.9); Timegroup = 3 (6.9, 11.1); and Timegroup = 4 (11, 233)) and time (in days) to URTP (Timegroup = 1 (2.7, 8.7); Timegroup = 2 (8.7, 12.7); Timegroup = 3 (12.7, 19.1); and Timegroup = 4 (19.1, 457)); thereafter, we included a strata interaction term based on the plotted residuals. The proportional hazards assumption was upheld in the overall model and across all covariates in the iRTP protocol duration multivariable cox proportional hazards model.

Effect estimates with 95% CI not including 1.00 were deemed statistically significant at *alpha* = 0.05. Data management and analyses across all three studies were conducted using R Studio v.4.1.0 (RStudio Inc., Boston, MA).

RESULTS

Race and SES among the NCAA–DoD CARE Consortium

Among the 22,819 SA included in these analyses, most participants were male (59.3%), White (75.0%), and from civilian institutions (82.9%) (Table 1a). Compared with female SA, a higher proportion of male SA were Black (17.4% vs 8.3%). Compared with male SA, a higher proportion of female SA were White (80.3% vs 71.3%). The distribution of race by NCAA SA type (i.e., civilian vs MSA cadet) was comparable. More than half of all SA reported a household income of >$120,000 (~53%), and 25% of the SA reported a household income over $210,000. For both sex and NCAA athlete, prevalence estimates across levels of median household income were comparable.

Race by NCAA primary sport. Across all sports, Asian SA were most prevalent in men's tennis (13.0%), women's tennis (12.3%), and women's diving (10.9%; Fig. 2). Black SA were most prevalent in men's basketball (42.1%), men's football (32.3%), and women's basketball (31.5%). White SA were most prevalent across all sports, particularly men's ice hockey (97.3%). The proportion of Black (42.1%) and White (44.6%) SA in men's basketball was comparable; conversely, White (54.0%) SA comprised a greater proportion of women's basketball compared with Black (31.5%) SA.

Race, NCAA primary sport, and household income. Figure 3 presents the average reported household income by race and sport. Across all sports, most White SA reported a median household income of between $120,000 and 150,000 or $150,000 and 180,000. Approximately 27% of Black SA reported a household income over $120,000 and 10% over $210,000. By comparison, nearly 60% of White SA reported a household income over $120,000 and a third (29%) reported a median household income greater than $210,000. Conversely, nearly 41% of Black SA reported a household income less than $60,000, whereas less than 10% of White SA reported a similar household income. By race, comparable proportions of self-reported median household income of $90,000–120,000, and $180,000–210,000 were observed. Table 1 presents the distribution of reported household income by race.

On average, SA who participated in men's basketball and football reported the lowest median household income of between $90,000 and 120,000. Compared with all other sports, SA in men's football (10.5%) and men's basketball (10.3%) had the highest prevalence of a household income of $0–30,000. Nearly half of SA from men's lacrosse (43.5%), women's lacrosse (39.9%), and men's water polo (37.4%) reported a median household income of >$210,000. A complete listing of sports and household incomes is presented in the supplemental materials (see Supplemental Fig. 1, Supplemental Digital Content, http://links.lww.com/MSS/C889).

Time to RTL

The final dataset evaluating RTL included 1724 athletes with SRC. Most SA identified as male (56.9%), in division I (75.2%), played a contact sport (75.4%), and did not have a history of SRC (61.1%; Table 1b). Across the sample, 70.71% of SRC occurred among White SA, 18.44% among Black SA, and 10.85% among non-Black/non-White SA.

Univariate injury and recovery characteristics by race. The mean time to RTL was 5.10 d (SD = 7.99;

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TABLE 1A. Descriptive statistics for each project—initial baseline race and SES characteristics in NCAA athletes.

	Overall	White	Black	Asian	Hawaiian/Pacific Islander	American Indian/Alaskan Native	Multiracial
Overall (*N* = 22,819)		17,115 (75.00%)	3132 (13.73%)	631 (2.77%)	134 (0.59%)	110 (0.48%)	1697 (7.44%)
Sex							
Female (*N* = 9290)		7463 (80.33%)	772 (8.31%)	316 (3.40%)	37 (0.40%)	43 (0.46%)	659 (7.09%)
Male (*N* = 13,529)		9652 (71.34%)	2360 (17.44%)	315 (2.33%)	97 (0.72%)	67 (0.50%)	1038 (7.67%)
NCAA athlete							
Civilian varsity (*N* = 18,913)		14,125 (74.68%)	2718 (14.37%)	509 (2.69%)	110 (0.58%)	96 (0.51%)	1355 (7.16%)
MSA varsity (*N* = 3906)		2990 (76.55%)	414 (10.60%)	122 (3.12%)	24 (0.61%)	14 (0.36%)	342 (8.76%)
Household income							
$0–30,000	1185 (5.19%)	434 (36.62%)	562 (47.43%)	30 (2.53%)	30 (2.53%)	15 (1.27%)	114 (9.62%)
$30,001–60,000	2329 (10.21%)	1261 (54.14%)	715 (30.70%)	73 (3.13%)	24 (1.03%)	27 (1.16%)	229 (9.83%)
$60,000–$90,000	3308 (14.50%)	2327 (70.34%)	580 (17.53%)	73 (2.21%)	26 (0.79%)	23 (0.70%)	279 (8.43%)
$90,000–$120,000	3952 (17.32%)	3065 (77.56%)	434 (10.98%)	99 (2.51%)	25 (0.63%)	15 (0.38%)	314 (7.95%)
$120,000–$150,000	2604 (11.41%)	2107 (80.91%)	241 (9.25%)	71 (2.73%)	6 (0.23%)	12 (0.46%)	167 (6.41%)
$150,000–$180,000	1909 (8.37%)	1539 (80.62%)	155 (8.12%)	69 (3.61%)	8 (0.42%)	4 (0.21%)	134 (7.02%)
$180,000–$210,000	1793 (7.86%)	1466 (81.76%)	145 (8.09%)	50 (2.79%)	5 (0.28%)	5 (0.28%)	122 (6.80%)
>$210,001	5739 (25.15%)	4916 (85.66%)	300 (5.23%)	166 (2.89%)	10 (0.17%)	9 (0.16%)	338 (5.89%)

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median = 3.00; IQR = 0.00, 7.00; Table 2a). The Kaplan–Meier survival analyses yielded several characteristics associated with time to RTL related to race (Supplemental Table 1, Supplemental Digital Content, Kaplan–Meier survival analysis estimates for time to RTL by race, http://links.lww.com/MSS/C889). Overall, SA who identified as Black were RTL within 2 d following injury (median = 2.0 d; 95% CI = 2.0, 2.0), in contrast to those who identified as White (median = 3.0 d; 95% CI = 3.0, 3.0) or as non-Black/non-White (median = 2.5 d; 95% CI = 2.0, 4.0) (Fig. 4A).

Multivariate predictors of time to return to normal academic performance. Among SA whose median difference in SCAT3 severity score was less than 20, there were no differences in time to RTL across race and the other evaluated variables (all *P* values >0.05), with the exception of SA who took longer than 7 d to become asymptomatic (Table 3). Indeed, their risk of RTL was 65% lower than those who became asymptomatic within a week following injury (HR_{Adj} = 0.35; 95% CI = 0.29, 0.42).

Furthermore, no differences by race were noted regarding predictors of risk of RTL among SA whose median difference in SCAT3 severity score was equal or greater than 20 (all *P* values >0.05), with the exception of sex (*P* = 0.001) and time to become asymptomatic (*P* < 0.001) (Table 3). Specifically, male SA were observed to have a 31% increased (HR_{Adj} = 1.31; 95% CI = 1.11, 1.53) risk of RTL compared with female SA, and SA who took longer than 1 wk to become asymptomatic experienced 65% decreased risk of RTL (HR_{Adj} = 0.35;

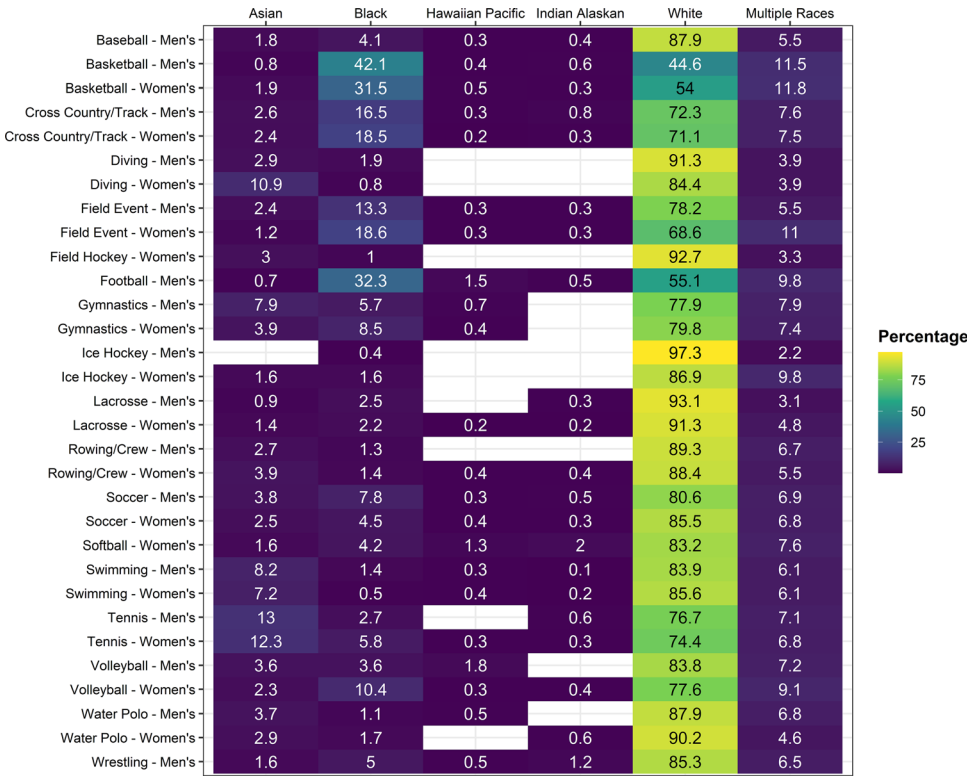

FIGURE 2—Race by primary NCAA sport. Shading indicates the percentage of athletes within a given racial category and sport.

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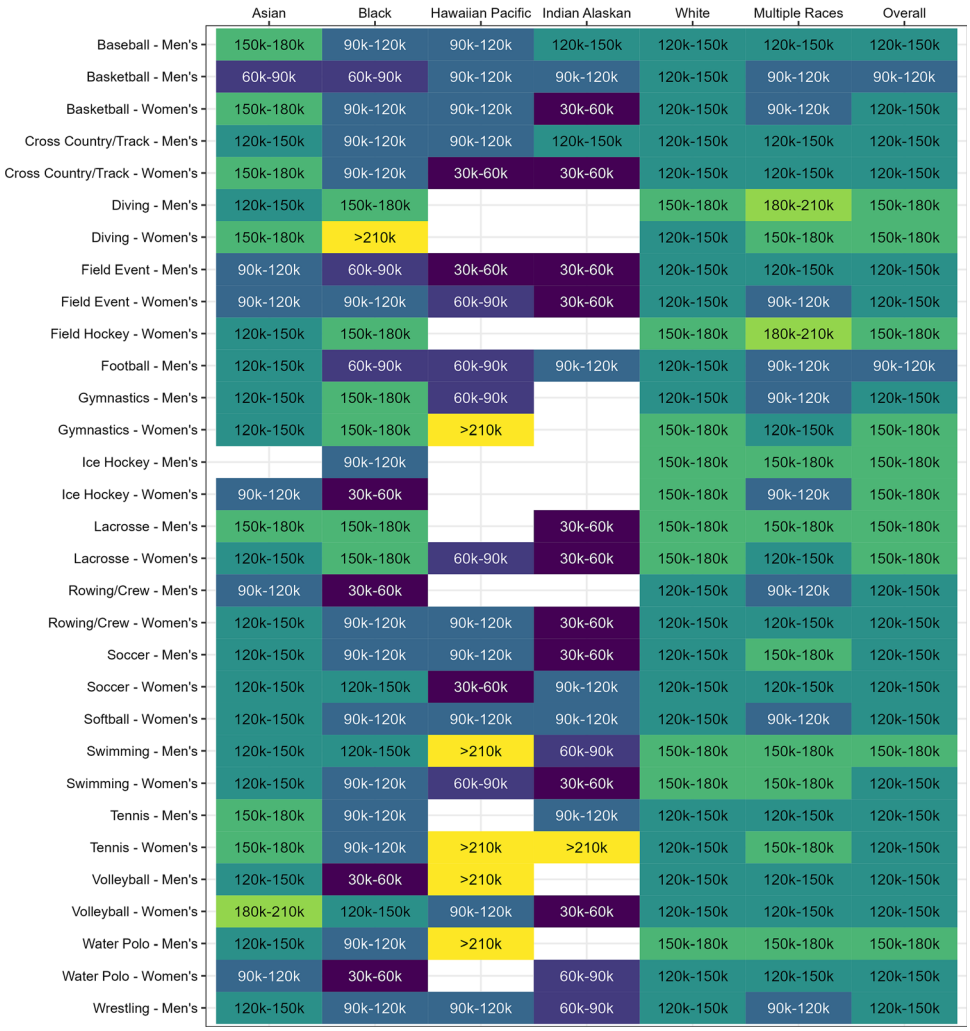

FIGURE 3—Average median household income by NCAA sport. Color shading is reflective of the relative median household income.

95% CI = 0.30, 0.40) compared with SA who took less than 1 wk to become asymptomatic.

Post-SRC academic supports by race. The overall distribution of post-SRC academic supports is provided in Table 2a. Approximately 54% of SA received academic support following SRC. Of the SA who received academic supports, most reported they were excused from class (74.74%), followed by extension for examinations/quizzes (55.50%) and homework (54.81%). Stratified by race, a greater proportion of SA who identified as White (56.37%) or non-Black/non-White (50.88%) received academic supports in contrast to those who identified as Black (46.96%).

Time to iRTP Protocol, Duration of RTP Protocol, and Time to URTP

Baseline and injury characteristics. The final dataset included 2334 SRC to evaluate iRTP, 2083 to evaluate RTP duration, and 2267 to evaluate URTP time (Fig. 1). Overall, a majority of SA with SRC identified as White (68.4%) and male (61.3%), and participated in a contact sport (77.6%; Table 1c). In addition, within each race, approximately 55% of the sample identified as males. The distribution of sex (χ^2 = 121, P < 0.001), sport contact category (χ^2 = 91, P < 0.001), and SRC history (χ^2 = 17, P = 0.01) varied across race. Prevalence of loss of consciousness following SRC was higher among Black SA (5.8%) than White (3.8%) and non-Black/non-White (2.1%; χ^2 = 7.4, P = 0.03). Postinjury (highest value within 48 h) SCAT3 symptom severity scores were comparable across race (χ^2 = 2.8, P = 0.20; Table 2b). Across the entire cohort, average days to iRTP protocol was 9.6 (SD = 11.9; median = 6.9; IQR = 3.9, 11.1, P < 0.001). The RTP protocol duration for all SA averaged 8.3 d (SD = 13.2; median = 5.8; IQR = 3.8, 8.2), with no differences observed across race (χ^2 = 5.0, P > 0.05). Average days to URTP for all SA was 17.1 d (SD = 18.9; median = 12.7; IQR = 8.7, 19.1), with noteworthy differences across race observed (χ^2 = 16, P < 0.001).

Univariate predictors of days to iRTP, RTP protocol completion, and URTP across race. Across the

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TABLE 1B. Time to return to normal academic performance—baseline characteristics by race.

Characteristic[a]	Overall N = 1724	White n = 1219	Black n = 318	Non-Black/Non-White[b] n = 187	P[c]
Sex					<0.001
Female	743 (43.10%)	589 (48.32%)	78 (24.53%)	76 (40.64%)	
Male	981 (56.90%)	630 (51.68%)	240 (75.47%)	111 (59.36%)	
Age					0.02
Mean (SD)	18.96 (1.23)	18.93 (1.17)	19.11 (1.37)	18.91 (1.31)	
Median (IQR)	19 (18.00, 20.00)	19 (18.00, 20.00)	19 (18.00, 20.00)	18 (18.00, 20.00)	
Sport category					<0.001
Contact	1300 (75.41%)	884 (72.52%)	285 (89.62%)	131 (70.05%)	
Limited contact	283 (16.42%)	223 (18.29%)	26 (8.18%)	34 (18.18%)	
Noncontact	141 (8.18%)	112 (9.19%)	7 (2.20%)	22 (11.76%)	
NCAA division					<0.001
Division I	1296 (75.17%)	904 (74.16%)	262 (82.39%)	130 (69.52%)	
Division II	212 (12.30%)	141 (11.57%)	37 (11.64%)	34 (18.18%)	
Division III	216 (12.53%)	174 (14.27%)	19 (5.97%)	23 (12.30%)	
Concussion history					0.03
0	1053 (61.08%)	738 (60.54%)	210 (66.04%)	105 (56.15%)	
1	479 (27.78%)	342 (28.06%)	81 (25.47%)	56 (29.95%)	
2	134 (7.77%)	100 (8.20%)	17 (5.35%)	17 (9.09%)	
3+	58 (3.36%)	39 (3.20%)	10 (3.14%)	9 (4.81%)	
Baseline SCAT3 severity score					<0.001
Mean (SD)	4.90 (8.23)	5.09 (8.16)	3.94 (8.30)	5.28 (8.50)	
Median (IQR)	2.00 (0.00, 6.00)	2.00 (0.00, 6.00)	0.50 (0.00, 4.25)	2.00 (0.00, 5.75)	
Anxiety disorder	38 (2.20%)	30 (2.46%)	4 (1.26%)	4 (2.14%)	0.5
ADHD	124 (7.29%)	87 (7.21%)	28 (9.12%)	9 (4.84%)	0.2
Depression	72 (4.27%)	54 (4.52%)	8 (2.61%)	10 (5.41%)	0.2
Autism spectrum	5 (0.29%)	2 (0.17%)	3 (0.97%)	0 (0.00%)	0.06
504 plan	31 (31.00%)	20 (33.33%)	10 (30.30%)	1 (14.29%)	0.8
IEP	47 (45.63%)	28 (44.44%)	17 (51.52%)	2 (28.57%)	0.5
Special education	104 (6.03%)	63 (5.17%)	34 (10.69%)	7 (3.74%)	<0.001

[a]Values described as count (frequency) unless otherwise specified; values may not equate to 100% due to missingness/unknown.
[b]Non-Black/Non-White includes the following self-reported racial categories: Hawaiian Islander, Multiple Races, Asian, and/or Alaskan Indian.
[c]Pearson's Chi-squared test; Kruskal-Wallis rank sum test; Fisher's exact test.
ADHD, Attention Deficit Hyperactivity Disorder; IEP, Individualized Education Plan; IQR, interquartile range; SCAT3, Sport Concussion Assessment Tool (3rd edition); SD, standard deviation.

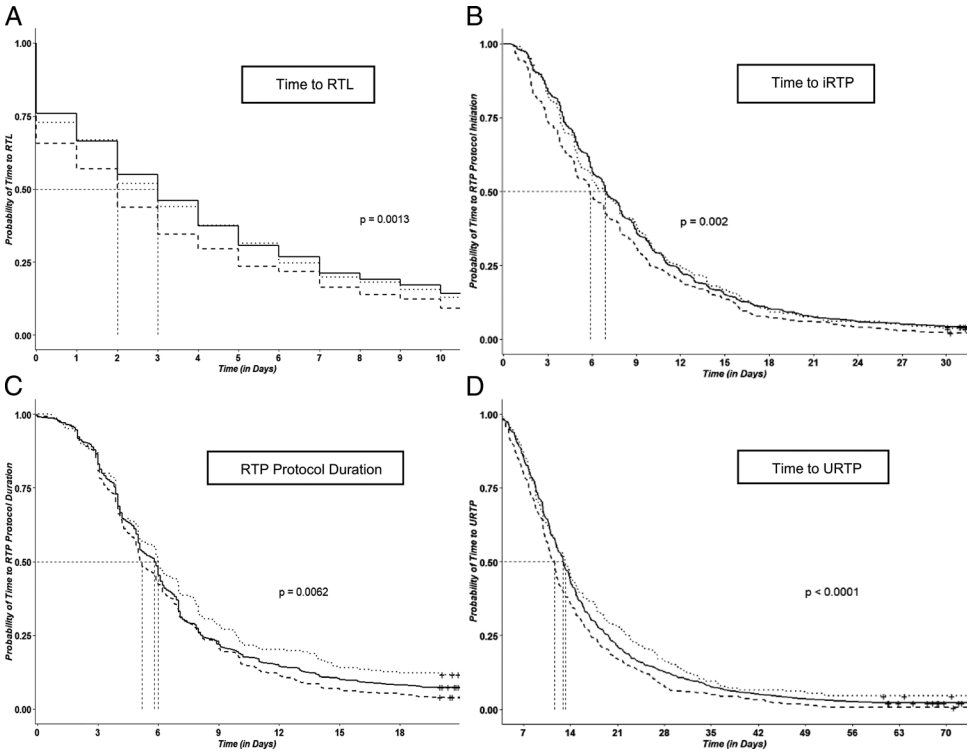

FIGURE 4—Kaplan–Meier curves for (A) days to return to normal academic performance (i.e., RTL); (B) days to iRTP protocol, (C) days to RTP protocol completion, and (D) days to URTP time across race (White, Black, non-Black/non-White). Log-rank tests were conducted to assess statistical difference in days to event for the three aforementioned SRC recovery endpoints. *Solid line*, White; *dashed line*, Black; *dotted line*, non-Black/non-White.

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TABLE 1C. Time to iRTP protocol, duration of RTP protocol, and time to URTP—baseline characteristics by race.

Characteristic[a]	Overall N = 2646	White n = 1810	Black n = 556	Non-Black/Non-White[b] n = 280	P[c]
Sex					<0.001
Female	1024 (38.70%)	814 (44.97%)	106 (19.06%)	104 (37.14%)	
Male	1622 (61.30%)	996 (55.03%)	450 (80.94%)	176 (62.86%)	
Age					0.015
Mean (SD)	18.94 (1.22)	18.91 (1.17)	19.09 (1.35)	18.86 (1.28)	
Median (IQR)	19 (18.00, 20.00)	19 (18.00, 20.00)	19 (18.00, 20.00)	18 (18.00, 20.00)	
Sport category					<0.001
Contact sport	2054 (77.63%)	1333 (73.65%)	512 (92.09%)	209 (74.64%)	
Limited contact sport	410 (15.50%)	333 (18.40%)	36 (6.47%)	41 (14.64%)	
Noncontact sport	182 (6.88%)	144 (7.96%)	8 (1.44%)	30 (10.71%)	
NCAA division					<0.001
Division I	2056 (77.70%)	1377 (76.08%)	477 (85.79%)	202 (72.14%)	
Division II	292 (11.04%)	193 (10.66%)	54 (9.71%)	45 (16.07%)	
Division III	298 (11.26%)	240 (13.26%)	25 (4.50%)	33 (11.79%)	
Concussion history					0.1
≤2	2557 (96.64%)	1750 (96.69%)	542 (97.48%)	265 (94.64%)	
≥3	89 (3.36%)	60 (3.31%)	14 (2.52%)	15 (5.36%)	
Baseline SCAT3 severity score					<0.001
Mean (SD)	4.72 (8.00)	4.95 (8.13)	3.79 (7.57)	5.08 (7.92)	
Median (IQR)	2 (0.00, 6.00)	2 (0.00, 6.00)	1 (0.00, 5.00)	2 (0.00, 6.00)	
Anxiety disorder	56 (2.12%)	45 (2.49%)	5 (0.90%)	6 (2.14%)	0.08
ADHD	215 (8.28%)	153 (8.55%)	47 (8.77%)	15 (5.51%)	0.2

Baseline data are of the SA's first baseline. Sport contact level was defined as previously published with contact sports including football, field hockey, wrestling, basketball, diving, ice hockey, lacrosse, soccer, and water polo; limited contact sports including baseball/softball, beach volleyball, fencing, gymnastics, volleyball, cross-country/track and field events; and noncontact sports including cheerleading, swimming, rowing/crew, golf, and tennis.

[a]Values described as count (frequency) unless otherwise specified; values may not equate to 100% because of missingness/unknown.

[b]Non-Black/non-White includes the following self-reported racial categories: Hawaiian Islander, Multiple Races, Asian, and/or Alaskan Indian.

[c]Pearson's x^2 test; Kruskal–Wallis rank sum test; Fisher's exact test.

ADHD, attention deficit hyperactivity disorder; IEP, individualized education plan.

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entire cohort, average days to iRTP protocol was 9.6 (SD = 11.9; median = 6.9; IQR = 3.9, 11.1; Table 2b). The Kaplan–Meier median days to iRTP differed significantly across race (P = 0.002; Fig. 4B). White SA entered the RTP protocol at a median of 6.9 d (95% CI = 6.8, 7.3), Black SA entered the RTP protocol at a median of 5.9 d (95% CI = 5.3, 6.7), and non-Black/non-White SA entered the RTP protocol at a median of 6.9 d (95% CI = 5.9, 7.9). In males, days to iRTP were significantly different across race (P = 0.04; Supplemental Table 2, Supplemental Digital Content, Kaplan–Meier estimates for days to iRTP protocol, days to RTP protocol completion, and days to URTP by sex, NCAA sport type, concussion history, and SCAT3 symptom severity score, http://links.lww.com/MSS/C889).

The RTP protocol duration for all SA averaged 8.3 d (SD = 13.2; median = 5.8; IQR = 3.8, 8.2), with no differences observed across race (Table 2b). The Kaplan–Meier median days to RTP protocol completion differed across race (P = 0.006; Fig. 4C). White SA median days to RTP protocol completion were 5.8 d (95% CI = 5.4, 6.0), Black SA median RTP protocol duration was 5.2 d (95% CI = 5.0, 5.9), and non-Black/non-White SA median RTP protocol duration was 6.0 d (95% CI = 5.4, 7.0). No difference in RTP protocol duration was observed across race when stratified by sex or sport contact level (all P values >0.05; Supplemental Table 2, Supplemental Digital Content, http://links.lww.com/MSS/C889).

Average days to URTP for all SA were 17.1 d (SD = 18.9; median = 12.7; IQR = 8.7, 19.1), with differences across race observed (Table 2b). The Kaplan–Meier median days to URTP significantly differed across race (P < 0.001; Fig. 4D). Median days to URTP was 13 (95% CI = 13, 13) for White SA, 12 (95% CI = 11, 12) for Black SA, and 13 (95% CI = 12, 15) for non-Black/non-White SA. Days to URTP was significantly different in males (P = 0.007) but not females (P = 0.11) across race (Supplemental Table 2, Supplemental Digital Content, http://links.lww.com/MSS/C889).

Multivariate factors associated with iRTP protocol, RTP protocol completion, and URTP. Race (P = 0.09), sex (P = 0.12), and SRC history (P = 0.30) were not significantly associated with iRTP (Table 4). However, sport contact level and difference in SCAT3 symptom severity score significantly varied as a function of days to iRTP protocol. Race (P = 0.009), sex (P = 0.04), and difference in SCAT3 symptom severity scores (between postinjury and baseline scores; P < 0.001) were significantly associated with RTP protocol duration (Table 4). Non-Black/non-White SA were associated (HR_{Adj} = 0.83; 95% CI = 0.71, 0.97) with 17% decreased risk of RTP protocol completion compared with White SA. Lastly, race (P = 0.002) and sex (P = 0.024) were significantly associated with URTP (Table 4). Black SA were associated with 17% increased risk of (HR_{Adj} = 1.17; 95% CI = 1.05, 1.31) URTP completion compared with White SA.

DISCUSSION

Universities commonly strive for more racially diverse student bodies and are in a unique position to increase educational opportunities for historically marginalized groups. With increasing focus on the biopsychosocial approach to injury management, this investigation sought to elucidate the relationship between race and household income characteristics of NCAA SA within the CARE Consortium, along with post-SRC recovery timeline based on self-identified race.

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TABLE 2A. Injury and recovery characteristics by race—time to RTL.

Characteristic[a]	Overall N = 1724	White n = 1219	Black n = 318	Non-Black/Non-White[b] n = 187	P[c]
Loss of consciousness	75 (4.36%)	49 (4.03%)	21 (6.62%)	5 (2.67%)	0.065
Altered mental status	775 (45.27%)	539 (44.47%)	150 (47.77%)	86 (46.24%)	0.6
Worst SCAT3 severity score (<48 h)					0.019
Mean (SD)	29.58 (21.56)	29.72 (20.81)	27.41 (22.80)	32.20 (23.77)	
Median (IQR)	24 (13.00, 43.00)	24 (14.00, 42.00)	21.5 (9.25, 41.00)	27 (13.00, 49.00)	
Difference in baseline and postinjury SCAT3 severity scores					0.4
Mean (SD)	24.69 (21.84)	24.60 (21.26)	23.70 (23.10)	26.86 (23.29)	
Median (IQR)	20 (9.00, 38.00)	20 (9.00, 37.00)	18.5 (7.00, 37.00)	19.5 (8.25, 42.75)	
Difference in baseline and postinjury SCAT3 severity scores, *n* (%)					0.6
<20	803 (49.54%)	560 (48.82%)	152 (52.05%)	91 (50.00%)	
>20	818 (50.46%)	587 (51.18%)	140 (47.95%)	91 (50.00%)	
Time to RTL					<0.001
Mean (SD)	5.10 (7.99)	5.45 (8.69)	3.86 (5.53)	4.91 (6.37)	
Median (IQR)	3 (0.00, 7.00)	3 (1.00, 7.00)	2 (0.00, 5.00)	3 (0.00, 6.00)	
Time to RTL (quartiles)					0.006
0 d	453 (26.28%)	293 (24.04%)	109 (34.28%)	51 (27.27%)	
1–3 d	518 (30.05%)	365 (29.94%)	99 (31.13%)	54 (28.88%)	
4–7 d	406 (23.55%)	303 (24.86%)	58 (18.24%)	45 (24.06%)	
>8 d	347 (20.13%)	258 (21.16%)	52 (16.35%)	37 (19.79%)	
Time to asymptomatic (d)					<0.001
Mean (SD)	9.18 (9.33)	9.45 (9.62)	7.99 (7.89)	9.43 (9.54)	
Median (IQR)	6.8 (3.90, 11.00)	6.9 (4.10, 11.20)	5.8 (2.80, 9.85)	7 (3.90, 11.72)	
Time to asymptomatic					<0.03
<7 d	828 (51.88%)	573 (50.57%)	169 (58.89%)	86 (48.86%)	
>7 d	768 (48.12%)	560 (49.43%)	118 (41.11%)	90 (51.14%)	
Received any academic support[d]	863 (54.04%)	637 (56.37%)	139 (46.96%)	87 (50.88%)	0.01
Decreased homework	104 (12.05%)	83 (13.03%)	13 (9.35%)	8 (9.20%)	
Excused from class	645 (74.74%)	467 (73.31%)	107 (76.98%)	71 (81.61%)	
Extension for homework	473 (54.81%)	348 (54.63%)	79 (56.84%)	46 (52.87%)	
Extension for examinations/quizzes	479 (55.50%)	360 (56.51%)	71 (51.08%)	48 (55.17%)	
Extension for examinations/quizzes	110 (12.75%)	81 (12.72%)	17 (12.23%)	12 (13.79%)	
Note taker	45 (5.21%)	30 (4.71%)	10 (7.19%)	5 (5.75%)	

Clinical assessments examined were the most severe between the two if SA had those data collected at postinjury and 24- to 48-h time point.
[a]Values described as count (frequency) unless otherwise specified; values may not equate to 100% because of missingness/unknown.
[b]Non-Black/non-White includes the following self-reported racial categories: Hawaiian Islander, Multiple Races, Asian, and/or Alaskan Indian.
[c]Pearson's x^2 test; Kruskal–Wallis rank sum test; Fisher's exact test.
[d]Values may not equal 100% as SA may have received multiple types of academic supports (e.g., decreased homework and note taker).

Race and SES among the CARE Consortium SA

The findings from this investigation demonstrate that the samples of collegiate SA presented herein are predominately White (75%) and come from households with incomes substantially higher than the national average (85% over $60,000) (11). This finding sharply contrasts with more than 40% of Black SA who come from households of less than $60,000. Overall, SA enrolled in the CARE Consortium approximate the distribution of all NCAA SA (34), albeit with a higher percentage of White and lower percentage of Asian SA than the student body across the United States (35).

Participating in varsity athletics is one potential opportunity for SA from socioeconomically disadvantaged backgrounds to attend college and earn a bachelor's degree or other advanced degrees. Specifically, collegiate SA are likely to graduate at higher rates than nonathletes (36), and college degrees are known to increase lifetime earnings (37), one factor in improving overall SES. The present findings, however, highlight that a disproportionate number of SA enrolled within the CARE Consortium come from higher income households.

Indeed, racial and SES privileges are unique constructs with intertwined effects (38). SES is a well-documented social determinant of health that influences access, care, and overall experience with the medical system before arriving at a college campus (39). Additional research is warranted at the collegiate level to understand how SA race and SES relative to their medical providers may influence their willingness/likelihood to seek care and/or the quality of care they receive. These findings are of importance for a variety of reasons: 1) acknowledging the lack of racial heterogeneity among NCAA sports highlights potential inequitable access to sport before college; and 2) given what is already known about the relationship between race/SES and medical care (14,40–42), these factors should be accounted for in sports medicine curricula and practice.

Time to RTL

Kaplan–Meier estimations suggested Black SA RTL 2 d following injury, in contrast to those who identified as White (median = 3.0 d) or as non-Black/non-White (median = 2.5 d). After controlling for covariates, only time to become asymptomatic was found to be a significant predictor of time to RTL across the multivariate models, suggesting post-SRC RTL management is relatively comparable across the sample, irrespective of race.

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TABLE 2B. Injury and recovery characteristics by race—time to iRTP protocol, duration of RTP protocol, and time to URTP.

	Overall	White	Black	Non-Black/Non-White	
	N = 2646[a]	*N* = 1810[a]	*N* = 556[a]	*N* = 280[a]	*P*[b]
Injury setting, *n* (%)					0.4
Competition	904 (34.16%)	619 (34.20%)	181 (32.55%)	104 (37.14%)	
Practice/training	1742 (65.84%)	1191 (65.80%)	375 (67.45%)	176 (62.86%)	
Postinjury SCAT3 symptom severity score					0.2
Mean (SD)	30.36 (21.61)	30.41 (21.01)	29.71 (23.14)	31.30 (22.27)	
Median (IQR)	26.00 (13.00, 44.00)	26.00 (14.00, 43.00)	25.00 (11.00, 44.00)	27.00 (12.00, 46.00)	
Unknown	148	105	34	9	
Difference between postinjury and baseline SCAT3 symptom severity score					>0.9
Mean (SD)	25.64 (21.84)	25.42 (21.28)	26.12 (23.59)	26.06 (21.98)	
Median (IQR)	21.50 (9.00, 39.00)	22.00 (10.00, 38.00)	21.00 (7.00, 41.25)	21.00 (9.00, 42.00)	
Unknown	172	117	44	11	
Loss of consciousness, *n* (%)	106 (4.02%)	68 (3.77%)	32 (5.81%)	6 (2.14%)	0.025
Unknown	12	7	5	0	
Altered mental status, *n* (%)	1,234 (47.21%)	820 (45.84%)	277 (50.73%)	137 (49.10%)	0.11
Unknown	32	21	10	1	
Days to iRTP protocol					<0.001
Mean (SD)	9.62 (11.90)	10.11 (13.17)	8.01 (7.48)	9.49 (9.56)	
Median (IQR)	6.90 (3.90, 11.10)	6.90 (4.10, 11.20)	5.90 (2.95, 9.90)	6.90 (3.90, 12.00)	
Unknown	312	196	85	31	
Time to iRTP protocol, *n* (%)					
≤1 wk	1,232 (52.78%)	827 (51.24%)	277 (58.81%)	128 (51.41%)	0.014
≤2 wk	1,944 (83.29%)	1,341 (83.09%)	399 (84.71%)	204 (81.93%)	0.6
≤3 wk	2,168 (92.89%)	1,494 (92.57%)	444 (94.27%)	230 (92.37%)	0.4
≤4 wk	2,232 (95.63%)	1,535 (95.11%)	459 (97.45%)	238 (95.58%)	0.09
>4 wk	95 (4.07%)	74 (4.58%)	11 (2.34%)	10 (4.02%)	0.09
RTP protocol duration (d)					0.08
Mean (SD)	8.29 (13.23)	8.46 (14.50)	6.73 (5.11)	10.19 (15.08)	
Median (IQR)	5.80 (3.80, 8.20)	5.80 (3.90, 8.10)	5.20 (3.45, 8.10)	6.00 (3.90, 9.90)	
Unknown	563	377	129	57	
Days to URTP time					<0.001
Mean (SD)	17.06 (18.93)	17.46 (20.03)	14.69 (14.10)	19.26 (19.75)	
Median (IQR)	12.70 (8.70, 19.10)	12.90 (8.90, 19.62)	11.60 (7.85, 17.10)	13.25 (8.80, 22.42)	
Unknown	379	266	73	40	
Time to URTP, *n* (%)					
≤1 wk	363 (16.01%)	232 (15.03%)	99 (20.50%)	32 (13.33%)	0.008
≤2 wk	1,316 (58.05%)	874 (56.61%)	310 (64.18%)	132 (55.00%)	0.008
≤3 wk	1,793 (79.09%)	1,218 (78.89%)	402 (83.23%)	173 (72.08%)	0.002
≤4 wk	1,995 (88.00%)	1,349 (87.37%)	446 (92.34%)	200 (83.33%)	<0.001
>4 wk	253 (11.16%)	185 (11.98%)	31 (6.42%)	37 (15.42%)	<0.001

Clinical assessments examined were the most severe between the two if SA had those data collected at multiple time points postinjury within 24–48 h.
[a]Race: White, Black, and non-Black/non-White (Hawaiian Islander, Multiple Races, Asian, Alaskan Indian).
[b]Pearson's x^2 test; Kruskal–Wallis rank-sum test.

TABLE 3. Multivariable cox proportional hazards regression modeling time to RTL by race based on the difference in baseline and postinjury SCAT3 severity scores.

	Difference in Baseline/Postinjury SCAT3 Severity Scores (<20) (*n* = 732)			Difference in Baseline/Postinjury SCAT3 Severity Scores (≥20) (*n* = 771)		
	HR_{Adj}	95% CI	*P*	HR_{Adj}	95% CI	*P*
Race			0.13			0.60
White	—	—		—	—	
Black	1.23	1.01, 1.49		1.10	0.90, 1.34	
Non-Black/non-White[a]	1.05	0.83, 1.32		1.01	0.80, 1.27	
Sex			0.40			0.001
Female	—	—		—	—	
Male	1.07	0.91, 1.26		1.31	1.11, 1.53	
Sport category			0.20			0.13
Contact	—	—		—	—	
Limited contact	0.84	0.68, 1.03		1.07	0.87, 1.31	
Noncontact	1.01	0.73, 1.40		1.31	1.02, 1.68	
Concussion history			0.20			0.20
0	—	—		—	—	
1	0.98	0.83, 1.15		1.10	0.94, 1.30	
2	0.76	0.57, 1.00		0.81	0.62, 1.07	
3+	1.16	0.73, 1.82		1.13	0.76, 1.66	
Time to asymptomatic			<0.001			<0.001
<7 d	—	—		—	—	
>7 d	0.35	0.29, 0.42		0.35	0.30, 0.40	

Sport contact level was stratified as follows: contact sports included basketball, diving, ice hockey, lacrosse, soccer, and water polo; limited contact sports included baseball/softball, cheerleading, gymnastics, volleyball, cross-country/track and field events; and noncontact sports included swimming and tennis.
[a]Non-Black/non-White includes the following self-reported racial categories: Hawaiian Islander, Multiple Races, Asian, and/or Alaskan Indian.

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TABLE 4. Multivariable Cox proportional hazards model for each recovery time point (days to iRTP, days to RTP protocol completion, days to URTP).

Days to iRTP Protocol[a]	HR Adj	95% CI	P	Days to RTP Protocol Completion	HR Adj	95% CI	P	Days to URTP[a]	HR Adj	95% CI	P
Race			0.09	Race			0.009	Race			0.002
White	—	—		White	—	—		White	—	—	
Black	1.13	1.01, 1.27		Black	1.09	0.97, 1.22		Black	1.17	1.05, 1.31	
Non-Black/non-White[b]	1.03	0.90, 1.19		Non-Black/non-White[b]	0.83	0.71, 0.97		Non-Black/non-White[b]	0.89	0.77, 1.02	
Sex			0.12	Sex			0.039	Sex			0.024
Female	—	—		Female	—	—		Female	—	—	
Male	1.08	0.98, 1.19		Male	1.11	1.00, 1.24		Male	1.12	1.01, 1.23	
Sport contact category * Timegroup			<0.001	Sport contact category			0.2	Sport contact category * Timegroup			0.01
Contact sport * Timegroup = 1	—	—		Contact sport	—	—		Contact sport * Timegroup = 1	—	—	
Limited contact sport * Timegroup = 1	0.7	0.54, 0.90		Limited Contact sport	1.03	0.90, 1.18		Limited Contact sport * Timegroup = 1	0.87	0.67, 1.11	
Noncontact sport * Timegroup = 1	0.46	0.29, 0.74		Noncontact sport	0.86	0.71, 1.04		Noncontact sport * Timegroup = 1	0.49	0.30, 0.79	
SRC history			0.3	SRC history			0.2	SRC history			0.3
≤2	—	—		≤2	—	—		≤2	—	—	
≥3	0.87	0.68, 1.11		≥3	0.84	0.64, 1.09		≥3	0.87	0.69, 1.11	
Difference in SCAT3 Symptom Severity score[c] * Timegroup			<0.001	Difference in SCAT3 Symptom Severity score[c]			<0.001	Difference in SCAT3 Symptom Severity score[b] * Timegroup			<0.001
<21 * Timegroup = 1	—	—		<21	—	—		<21	—	—	
≥21 * Timegroup = 1	0.4	0.33, 0.48		≥21	0.82	0.75, 0.90		≥21	0.47	0.39, 0.56	
Sport contact category * Timegroup			0.022					Sport contact category * Timegroup			0.085
Limited Contact sport * Timegroup = 2	1.54	1.11, 2.14						Limited Contact sport * Timegroup = 2	1.08	0.77, 1.51	
Limited Contact sport * Timegroup = 3	1.15	0.81, 1.63						Limited Contact sport * Timegroup = 3	1.19	0.85, 1.67	
Limited Contact sport * Timegroup = 4	1.38	0.97, 1.96						Limited Contact sport * Timegroup = 4	1.08	0.76, 1.54	
Noncontact sport * Timegroup = 2	2.04	1.15, 3.61						Noncontact sport * Timegroup = 2	1.4	0.76, 2.59	
Noncontact sport * Timegroup = 3	2.02	1.14, 3.59						Noncontact sport * Timegroup = 3	2.2	1.24, 3.90	
Noncontact sport * Timegroup = 4	2.14	1.19, 3.82						Noncontact sport * Timegroup = 4	2.19	1.23, 3.90	
Difference in SCAT Symptom Severity score[c] * Timegroup			<0.001					Difference in SCAT Symptom Severity score[c] * Timegroup			<0.001
≥21 * Timegroup = 2	1.54	1.21, 1.96						≥21 * Timegroup = 2	1.28	1.00, 1.64	
≥21 * Timegroup = 3	1.92	1.50, 2.47						≥21 * Timegroup = 3	1.68	1.31, 2.16	
≥21 * Timegroup = 4	2.19	1.67, 2.86						≥21 * Timegroup = 4	2.04	1.57, 2.65	

Days to iRTP data were censored at 30 d, days to URTP from iRTP protocol data were censored at 20 d, and days to URTP data were censored at 60 d. Sport contact level was defined as previously published with contact sports including football, field hockey, wrestling, basketball, diving, ice hockey, lacrosse, soccer, and water polo; limited contact sports including baseball/softball, beach volleyball, fencing, gymnastics, volleyball, cross-country/track and field events; and noncontact sports including cheerleading, swimming, rowing/crew, golf, and tennis. The time-varying covariate, Timegroup, for the time (in days) to iRTP protocol model was parameterized accordingly: Timegroup = 1 (0.1, 3.9); Timegroup = 2 (3.9, 6.9); Timegroup = 3 (6.9, 11.1); and Timegroup = 4 (11, 233). The time-varying covariate, Timegroup, for the time (in days) to URTP model was parameterized accordingly: Timegroup = 1 (2.7, 8.7); Timegroup = 2 (8.7, 12.7); Timegroup = 3 (12.7, 19.1); and Timegroup = 4 (19.1, 457).

[a]Needed to time-varying covariate.

[b]Non-Black/non-White includes the following self-reported racial categories: Hawaiian Islander, Multiple Races, Asian, and/or Alaskan Indian.

[c]Difference in SCAT symptom severity score was derived as the difference between postinjury SCAT3 symptom severity score and baseline SCAT3 symptom severity score.

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Furthermore, the present analyses suggest the only predictor to affect time to RTL was time to become asymptomatic. Indeed, those who took longer than 7 d to become asymptomatic, regardless of their postinjury symptom severity, demonstrated a 65% increased risk of prolonged time to RTL. These findings are consistent with prior research suggesting initial symptom burden predicts SRC recovery timelines (43), in addition to recent work that sought to evaluate the relationship between initial cognitive symptoms and RTL duration, concluding those who endorse a higher cognitive ratio (i.e., cognitive symptom cluster score divided by the total post-SRC symptom scale) were at increased risk for prolonged RTL duration (44). Researchers further concluded a higher proportion of patients with a low cognitive symptom ratio achieved RTL by 7 d postinjury (82.2%), in contrast to those with a higher cognitive ratio score (69.9%) (44). Overall, these studies suggest SA who endorse acute cognitive symptoms may experience prolonged RTL, and therefore may benefit from a targeted rehabilitation program before returning to the classroom.

Time to iRTP, RTP Duration, and URTP

This study further investigated the relationship between race and SRC RTP among SA who participated in NCAA sports. Advancements in SRC evaluation and management guidelines have improved over recent years (25) and are now much more consistently applied SC; however, the present findings suggest race may be a factor in the time to complete the RTP protocol and total days to URTP. Black SA median days to iRTP (nonsignificant), days to RTP protocol completion, and days to URTP were all 1 d sooner compared with White and non-Black/non-White SA. The extent to which social factors (i.e., race, sport type, etc.) may influence SRC recovery is likely

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multifactorial. In addition, we found inconsistent sex-related differences across the RTP outcomes. Indeed, similar work conducted among Broglio et al. (5) concluded sex significantly influenced total duration of RTP, with males demonstrating quicker recoveries than females. However, when accounting for sex-comparable sports (e.g., soccer, basketball, etc.), Master et al. (45). concluded there are no sex-related differences in time to RTP Collectively, these findings suggest further work is warranted to investigate how social constructs, such as sex, influence recovery trajectories within the collegiate setting.

Furthermore, the majority of concussed SA herein participated in contact/collision sports (e.g., football), which carries a higher risk of injury than limited/noncontact sport SA, and individuals in contact/collision sports are known to recover at faster rates, which could explain these minor differences (46). Second, SRC knowledge disparities by race previously observed among NCAA SA could be a factor (46). A recent study among high school SA suggested more Black than White SA did not disclose their "bell-ringer" experiences because of competition (22). Third, pressure from teammates, family members, and/or coaching staff may cause the SA to minimize symptoms and potentially result in premature return to activity (47,48). Indeed, these aforementioned factors may exhibit a synergistic effect on the SA's desire to return to sport faster than they are physiologically ready to appease their teammates, coaches, and families (49). Because of limited number of teams, roster slots, and financial incentive of playing at the professional/national sport level, athletes may mitigate SRC symptoms to remain on the field. Moreover, this may be especially heightened among cohorts who grew up in economically disadvantaged households. Accordingly, these socioeconomic considerations may also further exacerbate an SA's unwillingness to disclose their injuries for fear of being removed from competition, and potentially, scholarship opportunities (50).

Although differences in SRC disclosure are likely multifactorial, they may be further explained by environmental contexts, including institutional racism (e.g., negative experiences from racially discordant healthcare providers) (22). In consideration of these factors, it is plausible that current strategies to enhance SRC education and retention may not be appropriate for all patient populations and therefore must be tailored to the population of interest. Specifically, prior work by Wallace et al. (13) has evaluated differences in SRC knowledge, reporting behaviors (22), and healthcare navigation between White and Black athlete populations (14). Although these contributions have made significant strides in considering the influence of race and other social determinants of health on concussion recognition and management, further work is warranted in the post-secondary setting. In addition, Kolberg et al. (51) recently used comic-based educational material to improve SRC education among adolescent populations and their parents/guardians. Researchers concluded this medium of dissemination contributed to the greatest increase in SRC knowledge and understanding in both groups (patients, parents/guardians) compared with standard medical discharge documentation. Therefore, it remains critical to conceptualize and develop educational materials that are tailored to specific social constructs (e.g., sport, age, setting, etc.) to improve disclosure behaviors and, overall, promote safe sports.

Limitations

Despite the novelty of this study, it is not without limitations. Race and ethnicity are multifactorial social constructs, and the data captured and evaluated here do not fully encapsulate the lived experiences of those identifying as an underrepresented minority. Ethnicity was not included in the analyses because of the low sample of Hispanic and/or Latinx SA in the CARE dataset, further limiting the generalizability of these findings. Furthermore, SES and social determinants of health are defined by several factors (e.g., race, ethnicity, median household income, education, occupation, air and water quality, etc.), all of which are not explicitly described in the present study, as the intent of the CARE Consortium was to characterize acute and long-term effects of SRC. Moreover, household income was based on self-report, which carries known limitations, and social stigma may have resulted in some SA choosing not to disclose, or misreport, their household income. Although CARE is one of the most comprehensive studies examining the acute effects of SRC in NCAA SA to date, it was not designed to investigate factors potentially driving intragroup and intergroup race-related differences, including social and structural determinants of health, including (but not limited to) SES. Additional research is warranted to investigate potential implicit and explicit biases in clinical care, and how these may influence SRC recovery trajectories, with the hopes to identify and implement strategies to address these issues in a systemic manner. Furthermore, most SRC captured in this dataset were from White SA. Although these numbers may be comparable to other investigations and the NCAA at large, these findings may not be relevant to the general population. Lastly, the analyses did not control for participation in specific sports (e.g., basketball vs football vs softball, etc.), which may have differing proportions across races.

CONCLUSIONS

This investigation is the first to report on the intersection of race and household income and SRC recovery in NCAA varsity SA using data collected by the NCAA–DoD CARE Consortium. The racial composition of varsity SA approximates that of the full NCAA SA population but has a greater proportion of White SA and lower proportion of Asian SA than the US population. Most notably, a substantial number (85%) of SA from this investigation report a household income in excess of the US median, whereas a disproportionate number (40%) of Black SA reported a household income below the median. Although time to RTL seemed to vary slightly by race, these findings are likely not statistically or clinically relevant and ultimately support the notion that SA of all races within this sample returned to the classroom on a similar timeline. However, additional findings do suggest race is a factor influencing the SRC recovery process, including days to RTP protocol completion and days to URTP, even after

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adjusting for several covariates. These findings support the need for future work that better elucidates biopsychosocial characteristics across the patient–provider dyad as it is critical to ensure equitable evaluation and management of SRC occurring at the collegiate level.

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Author contribution: A. J. B., A. K. M., and S. P. B. were involved in study conception and design, as well as the interpretation of data, statistical interpretation, manuscript drafting, and final approval. All remaining authors were responsible for study conception, design, and providing generous feedback on manuscript drafts. All authors read and approved the final manuscript.

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